



UniSens-E

The small yet powerful telemetry sensor measures
and transmits

Voltage / current / capacity / power / energy
/Brushless rpm / height / Climb

Specifications and operating instructions

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Introduction

The UniSens-E is a pure telemetry sensor specifically designed for all electric models. It is so small and light, but at the same time so powerful that it can be suitable for almost any model.

In accordance with our philosophy to support many systems, the UniSens-E also speaks the language of

Jeti Duplex (EX),
Multiplex M-Link
Graupner/SJ HoTT
Robbe/Futaba FASSTest S.BUS2
JR DMSS
FrSky
Spektrum (via the Spektrum-Adapter)
PowerBox P2Bus

The telemetry used in the UniSens-E must be specified, once only, in the settings. This is done either through our PC program "UniSens E-Tool" or the UniDisplay, or directly via a free channel of the remote control, see chapter 6.4. On delivery HoTT GAM is selected.

Without additional sensors the UniSens-E can measure the entire drive, ie voltage, current, capacity and even the rpm. A special feature is that a brushless rpm sensor is already built-in. For measuring rpm you just need to make a single connection to one of the three motor phases with the included cable.

Since a high resolution barometric (air pressure) level sensor is integrated, in addition to the height measurement a Vario is also achieved with the UniSens-E.

Similarly extensive alarms are programmable, and can be reported by telemetry to the respective transmitter.

The UniSens-E is available with various connector systems, so normally nothing needs to be soldered. The current measurement range is symmetrical in both directions. Thus as regards the pin assignment, it does not matter if the battery positive pole is male or female. The measurement direction is just simply set in the Setup.

With our UniDisplay, all values measured on the UniSens-E can be viewed live directly and all settings and alarms can be programmed.

Whether sailplane, aerobatic, helicopter, HLG or slow flyer, the UniSens-E can, due to its low weight and compact size be used in almost every area. Of course, the UniSens-E is not only suitable for model aircraft. It can be installed in RC boats, RC cars, etc.

2 What the UniSens-E can do

- complete measurement of electric drives with current, voltage, power, capacity, rpm and altitude
- integrated brushless rpm sensor for versions with 140 A and 280 A measuring range.
- full telemetry support for Jeti Duplex (EX), Multiplex M-Link, Graupner HoTT, Futaba S.BUS2, JR DMSS, FrSky, Spektrum and PowerBox P2Bus.
- height measurement with automatic zeroing after switching on
- barometric vario
- current measurement range up to 55 /140 / 280 A in both directions, voltage up to 60 V. (the maximum current carrying capacity of the connectors used must not be exceeded!)
- connection for receiver signal for the remote control of certain functions
- direct connection to the GPS logger is possible to record data in a file
- measuring the receiver battery voltage
- powered from the receiver battery
- status is indicated by LED
- direct viewing of live readings with our UniDisplay
- parameter setting possible via a PC, UniDisplay or telemetry.
- PC software for settings and updates (SM UniSens-E Tool) is available online at www.smmodellbau.de in the menu Software & Updates
- free firmware updates possible via the PC with our USB interface (Order No. 2550), or with an existing USB connection cable such as from Jeti, Multiplex or Graupner.
- due to its compact size and low weight can be used virtually anywhere

3. Technical Data

Current range **55A version**: 55 A in both directions, i.e. – 55A to + 55 A

Following durations are allowed:

- 40 A unlimited
- 50 A for 1 minute
- 55 A for 20 seconds

Depending on the plug-in system, however, the limitations of the plugs are much lower!

Reasonable values are as follows

- | | |
|----------------------------|-------------------------------|
| - XT30 | Continuous: 30 A / 20 s: 45 A |
| - 1.5 sq. mm Silicon cable | Continuous: 30A / 20 s: 45 A |

Current range **140A version**: 140 A in both directions, i.e. – 140A to + 140 A

Following durations are allowed:

- 100 A unlimited
- 120 A for 1 minute
- 140 A for 20 seconds

Depending on the plug-in system, however, the limitations of the plug are much lower!

Reasonable values are as follows:

- | | |
|---------------------------------------|---------------------------------|
| - MPX green (double contact) and XT60 | Continuous: 50 A / 20 s: 70 A |
| - XT90 | Continuous: 90A / 20 s: 120A |
| - 4 mm gold plated plug | Continuous: 80 A / 20 s: 100 A |
| - 5.5 mm gold plug | Continuous: 120 A / 20 s: 150 A |
| - 6.0 mm LMT gold plug | Continuous: 120 A / 20 s: 150 A |
| -4mm ² silicon cable | Continuous: 80 A / 20 s: 100 A |

Current range **280A version**: 280 A in both directions, ie – 280A to + 280 A

Following durations are allowed:

- 200 A unlimited
- 240 A for 1 minute
- 280 A for 20 seconds

Depending on the plug-in system, however, the limitations of the plug are much lower!

Reasonable values are as follows:

- | | |
|-------------------------------------|---------------------------------|
| - 6.0 mm LMT gold plug | Continuous: 140 A / 20 s: 160 A |
| -2 x 4mm ² silicon cable | Continuous: 160 A / 20 s: 200 A |

Voltage Range: 0 to 60 V

Altitude range: 0 to 8000 m above sea level, automatically zeroed at switch on

Receiver Voltage measurement: 3.8 V to 10 V

Data Rate: 10 Hz

Power supply: from receiver supply via the telemetry connection (from 3.8 V to maximum 10 V)

Power consumption: 25 mA from the receiver supply

External Connectors: 1 x connector for telemetry and power ("Link")
1 x phase connection for the brushless rpm measurement
1 x Servo signal input from the receiver
COM port for UniDisplay, GPS logger or PC

Dimensions: Yellow XT30	23 (34) x 23 x 8 mm
1.5 mm ² silicon cable	18 x 23 x 8 mm
Green MPX connector:	26 (38) x 29 x 9 mm
Yellow XT60 connector:	26 (50) x 29 x 9 mm
Yellow XT90 connector:	26 (60) x 34 x 11 mm
4mm gold connectors:	26 (55) x 22 x 9 mm
5.5 mm gold connectors:	26 (44) x 22 x 11 mm
6.0 mm LMT gold connectors:	26 (44) x 22 x 11 mm

Weight: 55 A version 4.5g – 7.5g
140 A version 10 g - 20 g
280 A version 17 g - 30 g
without telemetry cable (depending on plug-in system), telemetry cable 4g

4 Readings

The following measured values can be capture by the UniSens -E. But depending on the telemetry used only parts of it may be available at the transmitter.

Description	Unit	Content
Drive current	A	Current in amperes consumed by the drive, at up to 2 decimal places depending on the telemetry.
Drive voltage	V	Voltage of the drive battery, at up to 2 decimal places depending on the telemetry.
Capacity	mAh	Battery capacity consumed by the drive, with a fully charged battery it will be incremented from 0 mAh.
Energy	Wmin	Drive energy consumed in watt minutes.
Power	W	Power consumed by the drive in watts.
Speed	rpm	Speed of the brushless motor in rpm, for this a connection to a motor phase is needed: can also be translated into propeller speed by input of a gear ratio
Altitude	m	Height above the starting point, the height is measured by a barometric sensor. Before the start, the altitude is kept at about 0 m to compensate for drift due to air pressure changes.
Climb rate	m/s	Vario value from the barometric sensor
Receiver voltage	V or VRx	Voltage at the Rx connection
Servo pulse	us	Servo pulse width at the single connection at “Link”, may optionally be used to switch between Min/Live/Max values.
Air pressure	hPa	Air pressure measured by the barometric sensor.
Height gain	m	Height change in the last 10 seconds, recalculated every second and can be used to detect the trend when thermalling.
Remaining capacity	%	Residual charge of the drive battery calculated from the set alarm value for the capacity and the capacity consumed by the drive in mAh Example: Battery size 2200 mAh, alarm value set to 1760 mAh for 80% discharge ➔ Residual capacity is 100% when the battery is full and 0% when the 1760 mAh has been consumed.

5 Variants

The UniSens-E is available with various connector systems and in two measuring ranges. The basic module is identical, only the fixed soldered connector for connection of battery and controller are different.

The following variants are currently available with 55 A measuring range:



XT30 connector, No. 3107



1.5 mm² Silicon cable No.3108

Versions with 140A measuring range:



MPX plug green, No. 3100



XT60 connector, No. 3101



4mm gold connectors, No. 3102



5.5 mm gold plug, No. 3103



6.0 mm LMT plug, No. 3104



4mm² Silicon cable No.3105



XT90 connector, No. 3106

Versions with 280 A Measuring range:



6.0 mm LMT plug, No. 3114



2 x 4mm² Silicon cable No.3115

6 Operation of UniSens-E

6.1. Installation

Because of its light weight and compact design, the installation is straightforward. Since the UniSens-E will always be plugged directly between the battery and controller, it should not normally be specially secured.

6.2. Meanings of the LED

The UniSens-E has a red status LED

After switching on the power supply, a rapidly flashing LED shows the internal initialization.

In operation, there are the following signals

- :LED flashes according to the selected telemetry, see 6.4 Telemetry selection without PC
 - ➔ waiting, timed measurement has not yet started
- LED flashes uniformly
 - ➔ timed measurement was started by exceeding the current threshold

6.3. Basic Settings

The settings of the UniSens-E can be made with our software, "SM UniSens-E Tool" on the PC or Laptop, or alternatively with our UniDisplay or the Jeti and HoTT telemetry

The following settings are important so that the UniSens-E can measure correctly:

- **"Telemetrie Auswahl"** defines the telemetry used. From firmware v1.04 it must be specified here whether it will continue to automatically check for Jeti / HoTT / Multiplex, or if it is fixed and Futaba S.BUS2 or JR DMSS telemetry is specified.
- **"Fixed serial number"** so that Jeti and Futaba sensors can be interchanged.
- **"Strommessung"** chooses the direction of current flow, and so the positive current direction in which the power is measured.
When using the green MPX connector or the yellow XT60 connector choose "normal" so that positive currents are measured when discharging the Batteries.
For the gold connectors on drives with a socket on the positive terminal of the battery use "normal", if the positive terminal of the battery is a plug, then select "invers".
- **"Motorpole"** specifies the number of magnetic poles for the brushless speed measurement (typically inrunner 2 poles, outrunner 10 or 14 poles). If in doubt, this value must be obtained from the manufacturer, or even the magnets are counted.
- **"Getrieb"** is the gear ratio for the rpm measurement. With direct drive "1.00:1" must be set here. Otherwise the value of the gearbox specified.
- **"Vario Schwelle"** is the threshold for the vario signal via telemetry separated for rise and fall. Only when the climb / descent is greater than the threshold is a Vario tone generated by telemetry.
- **"Vario Ton"** defines whether the vario for climb / descent or both is active.
Here the Vario tone can be turned off completely.
- **"Vario factor"** specifies the factor by which the values are multiplied by the Vario for telemetry. Normally it is 1.0, so that the true m / s are displayed on the telemetry. In particular cases, by using a factor greater than 1 sound for the telemetry are made more sensitive, if the transmitter itself allows no such setting.

- **"VarioFilter"** allows an adaptation of the response speed of the varios
"Slow" corresponds approximately to the previous evaluation, "medium" and "fast" are correspondingly faster. At the same time, however, the signal of the varios is always more rapid.
- **"Höhe sanft Nullen"** keeps the altitude measurement at 0 m while the model is still on the ground. This suppresses a drift of the sensor, but also a change in air pressure. The altitude measurement on the ground is always at 0 m.
The detection of whether the model is on the ground is done via the vario. In some cases this standstill detection is activated in the air (extremely smooth flight on a slope), in which case the automatic zeroing of the altitude can be switched off here.
- **"Min/Max per Rx"** enables the selection of Live / Max / Min values in the telemetry display over a receiver channel on each individual orange cable
- **"Kapazität"** sets the mode for the capacity measurement: continued, always start at 0 mAh, or always continue. See below.
- **"Stromoffset"** specifies whether when you connect the drive battery to the UniSens-E any existing current flow should always calibrate to 0. Thus, the quiescent current of the servo etc. are hidden.

Capacity measurement options:

"fortgesetzt " (Continued) setting:

With the UniSens-E, a battery can be flown to empty in a series of flights. The UniSens-E remembers the consumed capacity (and energy) and starts again with this value unless a battery is connected at full voltage.

"immer bei 0 starten" (always start at 0) setting:

The discharged capacity is not saved and thus always starts from 0.

"immer weiter" (always continue) setting

The discharged capacity is stored and incremented with each new start. Here just a manual reset to 0 possible.

Manual reset of the capacity counter:

Start the UniSens-E and then turn off again during the rapid blinking (flickering) of the red LED

→ the counter goes back to 0 on the next boot

Or for manual reset of the min / max values by the receiver channel, see 8.2 Min and Max Values by Receiver Channel Switch.

6.4. Telemetry selection without PC

As of firmware v1.11, there is in addition to the PC software and the UniDisplay another option, the selection of the telemetry used without tools:

- For this purpose the Unisens E is connected to the receiver as described in section 8.2. The three-pole Telemetry Cable goes to the telemetry input of the receiver, the optional Rx cable goes to a free servo output.
- At the transmitter a control (switch or stick) is defined for this servo channel.
- In the first five seconds after turning on the Unisens E of this control is operated backwards and forwards at least 3 times.
- Then the LED goes out briefly and then flashes the telemetry selection always three times, in accordance with this list:

☐	Futaba	➔	1 x LED, pause
☐	JR DMSS	➔	2 x LED, pause
☐	HoTT GAM	➔	3 x LED, pause
☐	HoTT EAM	➔	4 x LED, pause
☐	HoTT ESC	➔	5 x LED, pause
☐	Jeti Ex	➔	6 x LED, pause
☐	M-Link	➔	7 x LED, pause
☐	FrSky	➔	8 x LED, pause
☐	HoTT Vario	➔	9 x LED, pause
☐	Spektrum	➔	10 x LED, pause
☐	PowerBox P2Bus	➔	11 x LED, pause

- ☐ For the desired selection of the encoder is moved twice more
➔ the LED goes on and the new selection is stored.
- ☐ If you switch the power back out before, no change is made.

6.5. Telemetry alarms

These alarms are output at the transmitter via the attached telemetry. Depending on the system a beep will sound and/or a warning by voice output. Please read the notes in the individual telemetry system's sections.

Once the model has landed, the acoustic output stops automatically, so until you turn off the model no disturbing messages will be heard.

- **"Strom"** (Current)
The alarm is active when the set current threshold is exceeded.
- **"startspannung"** (Start voltage)
This alarm is a warning before starting with a dead battery. As an example if 12.4 V is set for a 3s LiPo, the alarm is only active if an already flown empty battery is connected accidentally. A full 3s has approximately 12.6 V, which is significantly above the threshold.
- **"voltage"**
The alarm is active when the set voltage threshold is exceeded.
- **" Kapazität"** (Capacity)
The alarm is activated when the consumed capacity exceeds the preset value. The capacity Alarm is cleared after 10 seconds, but then comes permanently once an additional 5% of the set capacity is consumed.
This is the main alarm to protect the LiPos. An input of max. 80% of rated capacity should be set.
- **"Höhe"** (Altitude)
The alarm is active when the set level is exceeded. After 20 seconds the alarm is automatically cancelled and only reactivated when the height is exceeded again. Well suited for flying to a certain altitude with a tow plane or monitoring an altitude limit.
- **"Rx Spannung"** (Rx voltage)
For monitoring the receiver supply. The alarm is active when the voltage falls below the set voltage threshold.

- "Drehzahl min and max" (rpm)
To monitor the measured rpm. The alarm is active as long as the value exceeds or falls below the set value becomes.

6.6. Vario output via telemetry

The tone generation of the Varios is carried out in all cases (with the exception of the "old" Jeti telemetry without EX) in the telemetry transmitter or the telemetry box. For this purpose, the Vario value transmitted in UniSens-E is used in m / s. However, not all transmitters allow a setting of the sound output in order to hide certain areas or the sensitivity of the acoustic output.

Therefore, the UniSens-E has the following setting options, to influence the transmission of the variometer in all telemetry systems:

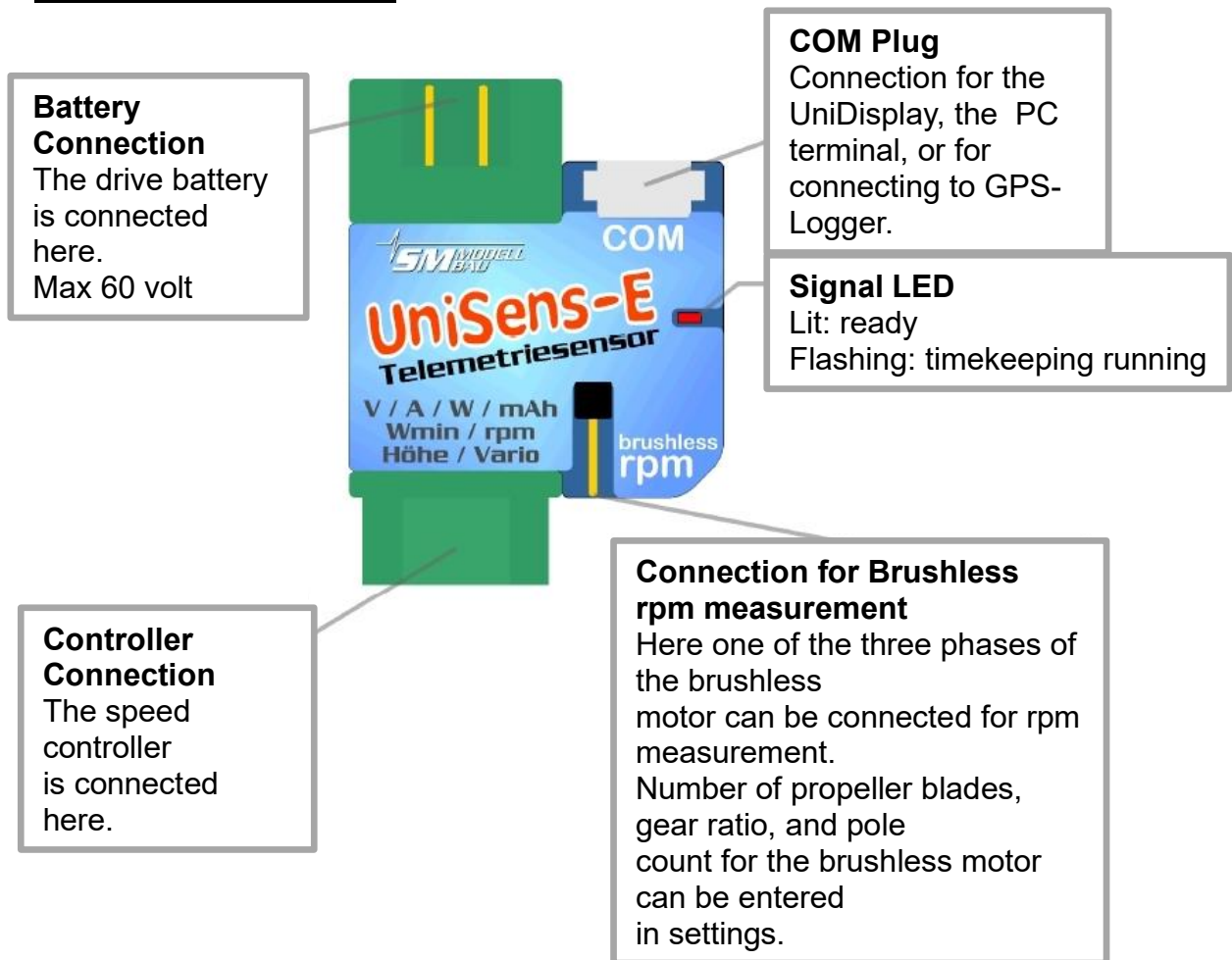
- **"Vario Schwelle"** specifies the response threshold for the vario signal via telemetry separately for ascending and descending. A vario tone is generated by telemetry only if the rise / fall is greater than the threshold.
- **"Vario Ton"** determines whether the Vario is active when rising / falling or both
Here the Vario tone can also be switched off completely.
- **"Vario Faktor"** determines the factor by which the values of the Vario are multiplied for the telemetry. Normally 1.0 is set here, so that the real m/s is displayed on the telemetry. In particular cases, however, with a factor greater than 1, the sound output of the telemetry can be made more sensitive. If the transmitter itself does not allow such a setting.
- **"Vario Filter"** allows an adaptation of the response speed of the vario.
"Slow" corresponds approximately to the previous evaluation, "medium" and "fast" are correspondingly faster. At the same time, however, the variable signal of the vario is always quicker.
- **"Vario bei Motor"** determines whether the Vario is active while the engine is running.

Example:

- "Vario Schwelle Steigen" (climb) is set to 0.5 m/s, "Vario Schwelle sinken" is set to -1.0 m/s
 - "Vario Ton" is set to "Auf" (up)
- ➔ if the model rises faster than 0.5 m / s, the value is transmitted
➔ if the model rises more slowly or sinks, 0 is transmitted by the vario

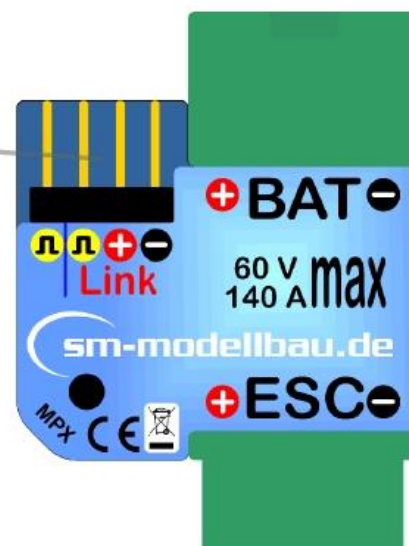
If the Vario value is always to be transmitted, the "Vario Schwelle" must be set to 0.0 m / s and "Vario Ton" is set to "auf/ab" (up / down).

7. Connections of UniSens-E



Telemetry Connector “Link”
Using the supplied patch cable a direct connection to the telemetry ports of the receivers is made here.
The cable and connection is identical for Jeti Duplex, Multiplex M-Link, and Graupner HoTT.
The UniSens-E is also supplied here with 3.8v to 10v.

With the single signal line, the Unisens E can be connected to a free channel of the receiver.
Over this channel from the transmitter you can switch the display of measured data between live, min and max values.



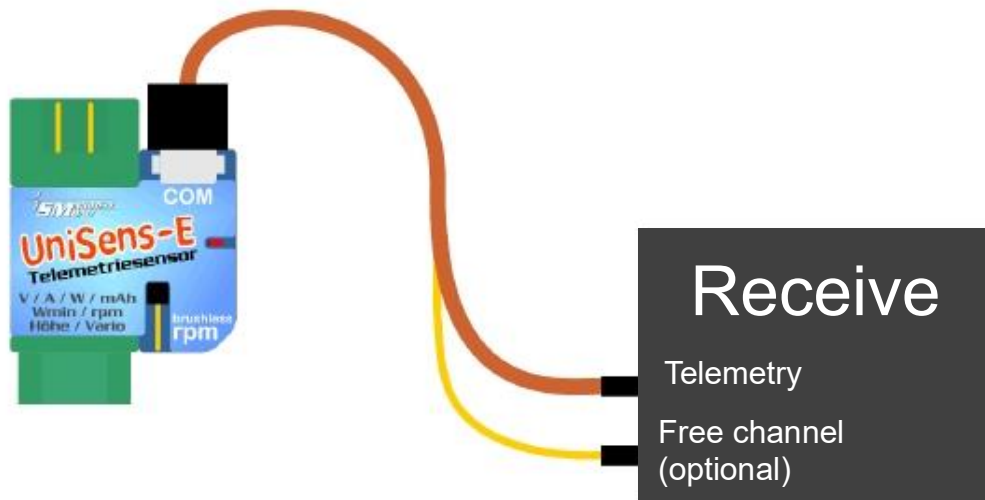
8. Connection Examples

Basically the UniSens-E is only powered over the telemetry connection "Link"

8.1. Only altimeter / vario e.g. in F3B/F3J sailplanes or HLG

For a pure altimeter and vario function via telemetry only connecting the UniSens-E to the receiver with the supplied cable is necessary. The altitude sensor with vario thus weighs only 10g plus cable.

On the UniSens-E, the cable is connected at "Link" and connected directly to the telemetry port. on the receiver.



8.2 Minimum and Maximum values switched via a Receiver channel

Optionally with a free receiver channel the Live, Maximum, and Minimum values can be switched between. To do this a second connecting cable is necessary between the desired receiver channel and the single pin on the UniSens-E telemetry connector.

On the transmitter a 3 position switch should be programmed for the selected channel, and the channel should change between the following values:

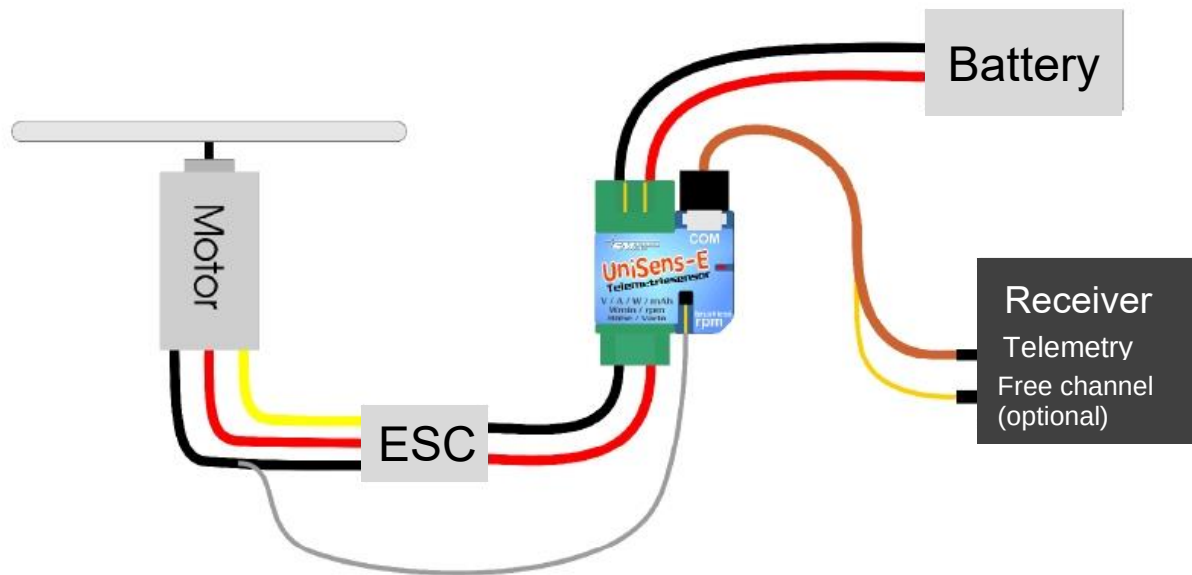
-100%	for Minimum values (change point 1.3 ms)
0%	for Live values
+100%	for maximum values (change point 1.7ms)

In addition the option "Min/Max per Rx" must be activated in the UniSens-E settings. If this option is not used the single receiver pin of the UniSens-E can just remain empty.

In operation, by a brisk three times back and forth of the 3 position switch on the transmitter the minimum and maximum values are reset.

If the capacity measurement is set to "immer weiter" (always continue), this reset of Min / Max also resets the capacity manually to 0. Prerequisite is that the model, after power on, was not yet in the air.

8.3. Electric power measurement



The UniSens-E is placed directly between the battery and controller. Since the UniSens-E is available with various connector systems, normally no soldering is required.

With the version with 4 mm or 5.5 mm and 6mm gold plated plug, and also the version with the silicon cable, only the positive terminal of battery and controller are connect directly to the UniSens-E. For the negative pole there is a short silicone cable. This separate negative pole only needs to be connected to the negative pole of the battery when the speed controller is equipped with an opto-coupler. Then the UniSens-E is missing the ground reference needed for measurement. With BEC regulators, the cable should be insulated with shrink tubing and just remain free.

The UniSens-E with a 5.5 mm or 6mm gold plated plug, and also the version with the silicon cable, does not have a plug on the single negative cable. Here a small connector should be attached with which the connection is made to the negative terminals of the battery. Well proven here is a 2 mm gold plated plug.

With the symmetrical current range of UniSens-E it does not matter how the connector on the battery is made. It is only necessary to set the correct current direction in the settings: When using the green MPX connector and yellow XT60 connector choose "normal" so that when the battery discharges positive currents are measured.

When using the bullet connectors for drives with the socket on the positive terminal of the battery choose "normal", with a plug on the positive pole of the battery select "invers".

When connecting the drive battery to the UniSens-E, depending on the settings for "Stromoffset" the current zero point is also calibrated. The current flowing then is set as 0 value.

8.4. Connecting the brushless rpm sensor

The brushless rpm sensor is built into the UniSens-E. To measure the rpm only a connection between one of the three motor phases and the single connection pin on the UniSens-E needs to be made.

This is a white single-pole cable.

This cable is easily cut to the required length, stripped to about 10 mm and plugged in one of the connections between Controller

and motor. Of course, the cable can also be soldered to a phase connection.



In the setup of UniSens-E for rpm measurement, the correct number of poles must be set for the brushless motor (a typical in-runner motor has 2 poles, an out-runner has 10 or 14 poles). When in doubt, this value must be requested from the manufacturer, or even the magnets (not stator!) counted.

In addition the gear ratio can also be specified, ie. if the rpm of an in-runner with gearbox is measured with the brushless speed sensor, a mounted gearbox can be include in the count. This gives the actual propeller speed.

8.5. Use of different plug systems

With the UniSense-E part No. 3105 and 3115 you have the possibility to use any plug. These variants are supplied with 4mm² cables.



Example of cabling for Part No. 3105 with Deans Ultra plug

The minus pole here is also only a thin single wire as only the positive pole goes through the UniSens-E. This negative lead must only be connected to the minus of the plug when the controller is equipped with an optocoupler. Then the UniSens-E lacks the ground reference point for the measurement. In BEC regulators, the cable should simply be insulated with shrink tubing and remain free.

If the connectors are soldered so that the battery is connected to the end of the UniSens-E where the "COM" and "link" are plugged in (as in the photo), then the current direction "normal" is chosen.

8.6. Connection to the GPS-logger

With the connection cables No. 2720 or 2721 the UniSens-E can be connected to, and communicate directly with, our Gps-Logger. The connection is made via the separate COM interface.

The GPS logger thus automatically records the data of the UniSens-E on its memory card. In This way GPS data and measurements of the UniSens-E are synchronised and can be evaluated together.

The connection of the UniLog 1/2 or UniSens-E only works with HoTT, Jeti and M-Link operation of the GPS-Logger 2. The COM interface cannot be used with the other telemetries.

The recording on the GPS logger is done with the storage rate of the GPS logger, ie a maximum of 10 Hz.

9 Using the UniDisplay

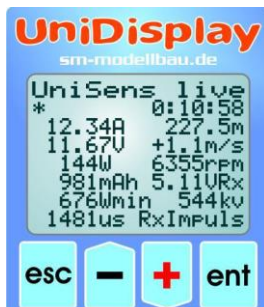
To connect the UniSens-E the firmware used in the UniDisplay must be at least v1.26. An update to the UniDisplay can be downloaded free from our website (www.sm-modellbau.de).

UniDisplay and UniSens-E are connected with the cable supplied with the display. The port used on the UniSens-E is marked with "COM". The connection cable can be connected either way, which end is the screen is irrelevant. The display is powered by the UniSens-E and automatically turns on when the UniSens-E is on. The display may be connected to the UniSens-E at any time.



Menu:

After switching on first the menu is activated. The menu options can be selected with the "plus" and "Minus" buttons, and the appropriate item selected with "Enter".



Live data display screen 1:

Here all current measured values are displayed.

Most values are self-explanatory.

- "Plus" starts and stops the recording.
- "Minus" alternates between live / MIN / MAX values.
- "Enter" toggles between the live screens 1, 2 and 3
- "Esc" will return to the menu.

Top right is the elapsed time.

The last line shows the measured servo pulse on the Rx port



Live display screen 2:

Here is the air pressure measured by the barometric pressure sensor, and the internal temperature of the UniSens-E.

Because of self-heating, this temperature always slightly higher than the ambient temperature.



Setup:

Here the menu appears for all of the settings UniSens-E.

The second line is the firmware version of UniSens-E and the Serial number.

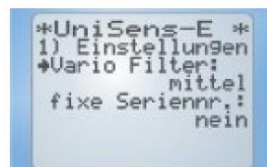
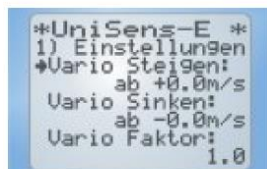
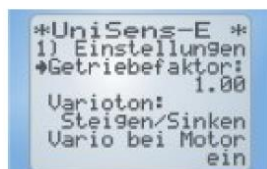
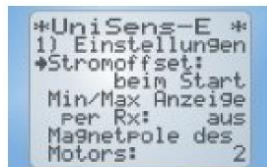
The menu options can be selected with the "plus" and "Minus" buttons, and the appropriate item selected with "Enter".



Settings screens:

The settings of the UniSens-E are summarized here.

The menu options can be selected with the “plus” and “Minus” buttons, and the appropriate item selected with “Enter”. The arrow then becomes a Dot and the selected value can be changed with “Plus” and “Minus”. A press on “Esc” or “Enter” stores the change.



- **Telemetrie:** default setting for the telemetry used.
- **HoTT Modus:** desired mode in HoTT.
- **Stromrichtung:** selects the measuring direction (sign) of the current sensor.
- **Kapaz.Messung:** enables or disables the continuous capacity measurement.
- **Stromoffset:** automatic zero point calibration of the current at power on or off.
- **Min/Max Anzeige per Rx:** allows you to switch between live/max /min values on the Telemetry display via a channel on the Rx.
- **Magnetpole des Motors:** specifies the number of poles for brushless rpm measurement
- **Getriebefaktor:** is the gear ratio for the rpm measurement
- **Varioton:** defines whether the Vario is active during climb / descent.
- **Vario Steigen:** is the positive threshold for the Vario signal.
- **Vario Sinken:** is the negative threshold for the Vario signal.
- **Vario Faktor :** Determines by which factor the values of the vario telemetry can be multiplied.
- **Vario Filter :** Allows the response speed of the varios to be adjusted
- **fixe Seriennummer** See Jeti and Futaba, allows the exchange of sensors on different models without having to redo all the settings.



Alarms:

Alarms can be set here for all telemetry versions. Depending on the telemetry alarms can also be specified directly on the transmitter. Please refer to the notes on the individual telemetry systems..

When the arrow is in the left column and the appropriate menu option activated with “Enter”, the value of the alarm can be set. After the arrow is moved right and the menu option activated with “Enter” the alarm can be activated (“+”) or deactivated (“-”) with “Plus” or “Minus”.



M-Link addresses:

For transfer of UniSens-E measured values by M-Link, addresses can be assigned here for the display on Multiplex transmitter. Each address may only be assigned once to any attached M-Link sensor, including the M-Link receiver. The Bus system ceases to function with multiple assignments.

If a value is not to be transmitted chose the address "--". This value becomes the highest permissible Address 15.



Jeti EX values:

Here, all measured values can be transferred to the Jeti transmitter or the JetiBox Profi and can be individually selected. Fewer measured values mean faster transmission, so you can hide the unneeded values here.



Spektrum Init:

The address specifies the assignment in the Spektrum adapter. Here is only important that with Several SM telemetry sensors on the Spektrum adapter no address is assigned twice. Normally the default is OK.

This includes all Spektrum sensors supported and operated by UniSens-E which can be individually selected or deselected. Each type of sensor may only occur once.



Futaba Init:

The compatibility mode for the Robbe Telemetry Box can be activated here.

10 Telemetry operation

The UniSens-E supports telemetry via Jeti Duplex (EX), Multiplex M-Link, Graupner HoTT, Robbe/Futaba S.BUS2, JR Propo DMSS, FrSky, Spektrum and PowerBox Core.

The telemetry operation is similar for all supported systems, live data is displayed on the transmitter or on external display, and with Jeti Duplex and HoTT the UniSens-E can also be operated from the transmitter. If the system output is a spoken voice, then this is also supported by UniSens-E.

The alarm output depends on the telemetry. In some systems, the UniSens-E generates the alarm, in others the thresholds are set directly on the transmitter. Please note the information on this below.

10.1. Jeti Duplex

The UniSens-E is a complete telemetry sensor for Jeti Duplex 2.4 GHz systems. The Jeti Expandere E4 for the connection of up to 4 sensors is supported.

Connection direct to the Jeti Duplex receiver is made with the patch cable supplied between “Link” on the UniSens-E and “ext” on the Jeti Duplex receiver. With the new REX receivers it must be ensured that the telemetry slot is configured on "JETIBOX / Sensor", or in the Central Boxes on "telemetry input". This setting can be made from the transmitter in the Device Explorer.

10.1.1. Ex Telemetry with the Jeti transmitter and the JetiBox Profi

The UniSens-E transfers the data using the Jeti EX telemetry to the DC/DS – 12/14/16/24 transmitter or the JetiBox Profi. These data values can be displayed on the transmitter, and voice outputs and alarms can be set. With the built-in emulation of the JetiBox the UniSens-E can be controlled and settings changed as described below.

If a UniSens-E is replaced by another, the telemetry values must be re-read in the DC/DS-14/16/24 or ProfiBox. Each UniSens-E has its own serial number and the Jeti EX distinguishes between the various devices on it.

Starting with firmware v1.11 the "fixed serial number" can be switched on. Each Unisens E then sends the serial number 12345 and the devices can be interchanged.

The following measured values are shown in Jeti EX operation

- | | | | |
|--------------|-------------------|----------------------|--------------------------|
| • Spannung | <i>Voltage</i> | • Energie | <i>Energy</i> |
| • Strom | <i>Current</i> | • Leistung | <i>Power</i> |
| • Kapazität | <i>Capacity</i> | • Luftdruck | <i>Air pressure</i> |
| • RxSpannung | <i>Rx Volts</i> | • Impuls ein | <i>Pulse in (servo)</i> |
| • Höhe | <i>Height</i> | • Temperatur intern | <i>Internal temp.</i> |
| • Vario | <i>Climb rate</i> | • Höhengewinn | <i>Height gain</i> |
| • Drehzahl | <i>rpm</i> | • Restkapazität in % | <i>Residual capacity</i> |

10.1.2. Operation of the UniSens-E with the JetiBox

< - Mx
v

After the start of transmission the JetiBox is changed to Mx for the attached sensors..

Firmware v1.17
SM UniSens-E

A press the on the ▼ key changes to the UniSens-E initialisation screen then the measured data is displayed.

A 23.28 V 221.8 m
36.04 A 1377 mAh

As soon as the first screen with measured data appears, the different data screens can be selected with presses of ◀ and ▶ keys. A pressure on the key ▲ starts the recording of data, which is indicated by an acoustic signal. A further pressure on ▲ terminates the recording.

A simultaneous long pressure on the keys ◀ and ▶ changes between the display of Live / MAX / MIN values.

In the top left of the screen there is an indicator of the current active data screen and/or the status of the UniSens-E:

:

A first data screen, following screens have B, C, etc

* recording running

> maximum values are indicated

< minimum values are indicated

< Strom Alarm >
(Aus) 50A

A press of key ▼ changes to the settings. Again with the keys ◀ and ▶ the different screens and the desired point are selected.

< Strom Alarm >
(Ein) < 49A >

After a further press of key ▼ the selected value can then be changed (keys ◀ and ▶). With a simultaneous pressure on ▲ and ▼ the alarm is switched on/off (Ein/Aus).

Changed settings are only stored with the move back to the selection level with ▲.

10.1.3. Display of measured values on the JetiBox

A 23.28 V 221.8 m 36.04 A 1377 mAh	Top: Drive voltage, barometric altitude from start point Bottom: Drive current, capacity used
B 1750.1 Wmin 2481 rpm 839 W	Top: e nergy used Bottom: rpm, drive power
C 5.01 VRx 221.8 m >>>>> _ +12.1 m	Top: receiver voltage, barometric altitude compared to start point Bottom: Graphical display of the Vario, Vario as numerical value
D 971.43 hPa internal 28.1 °C	Top: Current air pressure Bottom: internal temperature of UniSens-E. As a result of self-heating this temperature is always somewhat higher than the ambient temperature.
E 00:14:34	Top: measured time

10.1.4. Alarms

When operating on the Jeti transmitter modules with the display of data on the simple JetiBox all alarms and Vario sounds are generated directly from UniSens-E. All relevant settings are therefore made on the UniSens-E.

The JetiBox Profi and the Jeti transmitters in Jeti EX mode can actually generate the alerts and sounds. These are then set in the Box or in the transmitter. Alarms that are set in UniSens-E are additionally issued.

10.2. Multiplex M-Link

The UniSens-E is also a full telemetry sensor for the Multiplex M-Link 2.4 GHz system. The Measured values can be transmitted live to the ground and displayed directly on Multiplex Profi Tx, ROYAL pro or COCKPIT SX transmitters or the external telemetry screen displays.

In order to display the correct rpm, the Royal Pro transmitter must have at least firmware v3.46 and the external display at least v1.09.

The connection to the M-Link receiver is made with the Patch cable provided between the connection location “Link” on the UniSens-E and “Sensor” on the M-Link receiver.



The settings for telemetry can be made either with the UniDisplay (also see chapter 9) or with our “SM Uniens-E tool” software on a PC.

The addresses for display on the Multiplex remote control (the line that the respective value is indicated in) can also be freely selected.

10.2.1. Alarms

With M-Link all alarms are generated directly from the UniSens-E. All relevant settings are therefore made at the UniSens-E

There is a peculiarity with the Vario settings:

Since the Multiplex transmitter itself produces the Vario tone, the UniSens-E suppresses climb values which are smaller than “Vario threshold”. Thus this range is hidden from the transmitter tone.

Example: –“Vario threshold climbing” is set to 0,5 m/s

–“Vario sinking threshold set to -1.0 m/s

–“Vario tone” is set to “on”

➔ if the model rises faster than 0.5 m/s, the value is sent and the transmitters beeps

➔ if the model rises or sinks more slowly, the value 0 sent and the transmitter remains silent

If the change is always required, “Vario threshold” must be adjusted to 0,1 m/s and “Vario tone” set to “up/down”.

10.3. Graupner HoTT

The UniSens-E is also a full telemetry sensor for the Graupner HoTT 2.4 GHz system. The measured values can be transmitted live to the ground and displayed directly on either the Smart Box at the HoTT transmitter or directly in the display of the HoTT transmitter.

The connection to the HoTT receiver is made with the Patch cable provided, between the connection "Link" on the UniSens-E and "T" on the HoTT receiver.

The UniSens-E can be operated with HoTT either as a "General Module (GAM)", "Electric Air Module (EAM) "or" controller (ESC) ". The desired type is specified in the settings of UniSens-E. The preset default setting is GAM. This allows several UniSens-E to be operated together.

10.3.1. Alarms

The UniSens-E supports both the text mode and digital mode of the HoTT system. In both modes of operation all the adjustable alarms of the UnSens-E are indicated on the transmitter by beeping or voice output.

There is a peculiarity with the Vario settings:

Since the HoTT transmitter itself produces the Vario tone, the UniSens-E suppresses climb values which are smaller than "Vario threshold". Thus this range is hidden from the transmitter tone.

Example: - "

- "Vario climbing threshold" is set to 0.5 m/s, "sinking threshold" is set to -1.0 m/s
- "Vario tone" is set to "on"

→ if the model rises faster than 0.5 m/s, the value is sent and the transmitters beeps (for transmitters without built in speaker the vario sound will only be heard with headphones.

→ if the model rises or sinks more slowly, the value 0 sent and the transmitter remains silent
If the vario change is always required, "Vario threshold" must be adjusted to 0.0 m/s and "Vario tone" set to "up/down".

10.3.2. Vario

The HoTT transmitters display the range from -0.5 m / s to 0 m / s in the sound output. However, to obtain a continuous Vario tone, the UniSens-E skips this area.

For a more sensitive response of the acoustic output HoTT should have a "vario factor" of 2 to 4 set in the UniSens-E. The result is a good resolution acoustic output without a hole around the zero point. You have to accept, however, that the Vario value displayed and recorded in the transmitter does not correspond to reality.

10.3.3. Text Mode

To enter text mode, use the Telemetry menu and select "Settings view". With the left keypad of the transmitter using "On" and "Off", for UniSens-E call up the "General module".

With one click to the right, you leave the receiver data and display the text from UniSens-E.



Operation is done with the right touchpad on the transmitter. Structure and content are identical to the screens of the UniDisplay, see also Chapter 9. Here you can also set all the alarms, the beep from the transmitter or the voice output.

Operation in text mode seems to be rather slow because the data is not updated as often via telemetry.

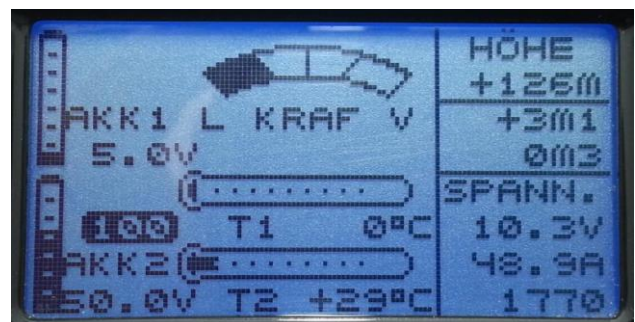
10.3.4. Digital mode

From the default display of the sender with the "left" and "right" of the left of the touchpad activate digital mode. Use the "On" and "Off" keys of left panel of the transmitter to call the correct HoTT mode, in accordance with the mode selected in the UniSens-E. With the "Left" and "Right" keys of the left touch pad you can now change between each screen. Depending on the mode there are additional screens available with a large display of measurements.

Display as GAM (General Air Module)

These values from the UniSens-E are allocated differently:

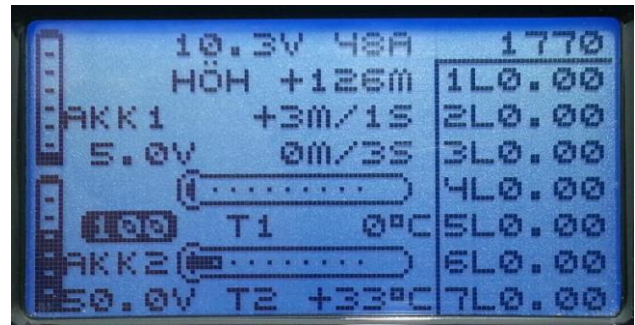
- Fuel scale: shows the remaining capacity of the battery according to the capacity set under "Alerts".
- Temperature 2: the internal temperature
- Battery 1: the receiver battery voltage.
- Battery 2: shows whether Min or Max values are selected by Rx Control: .
 - 0.0V minimum values
 - 50.0V live values are displayed
 - 99.9V maximum values
- **m3** shows the height gain in the last 10 seconds
- **m1** shows the energy in Wmins



Display as EAM (Electric Air Module)

These values from the UniSens-E are allocated differently:

- Temperature 2: the internal temperature
- Battery 1: the receiver battery voltage.
- Battery 2: shows whether Min or Max values are selected by Rx Control: .
 - 0.0V minimum values
 - 50.0V live values are displayed
 - 99.9V maximum values
- **m3** shows the height gain in the last 10 seconds



Display as an ESC (controller)

These values from the UniSens-E are allocated differently:

- Temperature : the internal temperature

In this mode Height and Vario cannot be displayed.



10.4. Futaba S.BUS2

Like any other sensor, the UniSens-E is connected to the S.BUS2 slot of the receiver.

Currently the UniSens-E is not yet registered in the transmitters, so it uses already existing sensor protocols. We tested the integration with the T14SG firmware v2.0, the FX-32 firmware v1.1 and the T18MZ Firmware v2.4.0 on the receivers and R7008SB R7003SB. Older firmware versions support the integration but possibly incomplete.

With S.BUS2 Servo data sensor values can be connected to the same data line. But since the servo data is far more important than the sensor values we strongly recommend that you make a strict separation. All servos go to the S.BUS1 connection of the receiver, all the sensors on the S.BUS2. Thus, in the event of an error, a sensor can never interfere with the data for the servos.

If nevertheless the UniSens-E is to be operated together with the servos on S.BUS2, is absolutely mandatory that a connection cable No. 9110 is used between UniSens-E and S.BUS2! Thus the sensor is decoupled from the bus so far that any influence on the servo data is impossible.

If a UniSens-E is exchanged for another, the telemetry values must be re-registered in the Futaba transmitter. Each UniSens-E has its own serial number and the Futaba system uses this to distinguish between the individual devices.

From firmware v1.11 onwards, however, the option "fixed serial number" can be switched on. Each UniSens-E sends the serial number 12345 and the units can be exchanged if they are registered identically in the transmitter.

If the option "fixed serial number" is changed, the UniSens-E will not work again until it has been re-registered with the transmitter.

10.4.1. Registration on the Transmitter

To use the UniSens-E with the S.BUS2, it must like all S.BUS2 sensors be first registered on the transmitter. For this purpose the "Link" connection of the UniSens-E is connected to the "S.I / F" socket on the transmitter. Some transmitters, such as the T14SG, do not provide power on this jack, in this case the UniSens-E can be supplied with a separate battery via a Y cable. The UniSens-E behaves like a Robbe / Futaba sensor and is thus closely integrated into the system. Please also refer to the transmitter instructions.

However, the UniSens-E currently uses six Robbe / Futaba sensor values to represent all measured values.

Special feature in the 280 A version:

The UniSens-E in the 280 A version sends Futaba the current value divided by 10! The reason for this is that the Futaba transmitters can display a maximum of 150A. In order to be able to represent the full range, e.g. for true 234 A, a value of 23.4 A is displayed. The capacity display is not affected by this and always shows the real value.

10.4.2. Robbe Telemetry Box

The Robbe Telemetry Box as a further possibility of the presentation of the Futaba telemetry data unfortunately does not recognise the temperature sensor SBS-01T used by UniSens-E. Therefore there is the option "Telemetry Box" In the settings of UniSens-E, which registers all Futaba SBS-01T sensors through the sensor TEMP125. Thus the UniSens-E can also be used with the Telemetry Box without major restrictions.

The example of the T18MZ here follows the steps of the application:

If the UniSens-E connected to the transmitter and is supplied with power, it is invoked in the sensor menu item "Login". In this way the sensor is registered in the transmitter and assigned free slots. The sensor and transmitter save this assignment.

Sensor			Test UniLog 2			59%			2/2
Zeitschlitz	Sensor	ID	Zeitschlitz	Sensor	ID	Zeitschlitz	Sensor	ID	
19	Funktion inaktiv		25	Funktion inaktiv		31	Funktion inaktiv		
20	Funktion inaktiv		26	Funktion inaktiv					erneut lesen
21	Funktion inaktiv		27	Funktion inaktiv					Neu anordnen
22	Funktion inaktiv		28	Funktion inaktiv					Anmeldung
23	Funktion inaktiv		29	Funktion inaktiv					Wechsle Zeitschlitz
24	Funktion inaktiv		30	Funktion inaktiv					

To be able to represent all values the menu item "Login" on the transmitter must necessarily be called 6 times. The message "OK" will appear four times, at the fifth time the message is "sensor already exists".

When the application of all sensors is complete, the sensor list looks like this:

In T18MZ the sensors can then be renamed.

Sensor			Test UniSens-E			55%			1/2
Zeitschlitz	Sensor	ID	Zeitschlitz	Sensor	ID	Zeitschlitz	Sensor	ID	
1	VARIO-F1712	14413	7	Funktion inaktiv		13	Funktion inaktiv		
2	VARIO-F1712		8	Funktion inaktiv		14	Funktion inaktiv		
3	CUR-F1678	14413	9	Funktion inaktiv		15	Funktion inaktiv		
4	CUR-F1678		10	Funktion inaktiv		16	Funktion inaktiv		
5	CUR-F1678		11	Funktion inaktiv		17	Funktion inaktiv		
6	Drehzahl	14413	12	Funktion inaktiv		18	Funktion inaktiv		

9 slots are occupied by 6 UniSens-E sensors:

Sensor	Name	Slots	Original designation	In UniSens-E	Value examples
1	VARIO-1712	2	Height	Hohe	155 m
2	CUR-F1678	3	Vario	Vario	13.2 m/s
			Current	Strom	48.4 A
			Voltage	Spannung	20.3 V
3	SBS 01RM / O	1	Capacity	Kapazität	3612 mAh
			U/min	Drehzahl	9284 rpm
4	SBS-01T	1	TEMPERAT	Energie	Wmin
5	TEMP 125	1	TEMPERAT	Residual capacity as a value	0 - 100 (%)
6	TEMP 125	1	TEMPERAT	Reception quality from the receiver data	Steps 25/50/75/100

Now connect the UniSens-E to the receiver and call up the transmitter telemetry display.

Here again the T18MZ for example.

Telemetrie		Test UniSens-E	
Empfänger	Extern	3. CUR-F1678(Strom)	6. Drehzahl(Magnet.)
5.1V	0.0V	48.4A	9264rpm
1. VARIO-F1712(Höhe)		3. CUR-F1678(Spannu...)	
155m		20.4V	
1. VARIO-F1712(Vario)		3. CUR-F1678(Kapazität)	
13.2m/s		3612mAh	

See the same values in the T14SG as follows (here the names cannot be changed):

TELEMETRIE		Til 1/2
Rx-BATT.	1 HÖHE	
5.0V	+155 m	
EMPFÄNGER	VARIO-1712	
EXT-VOLT	1 VARIO	
0.0V	+13.2m/s	
EMPFÄNGER	VARIO-1712	

TELEMETRIE		Til 2/2
3 STROM	3 KAPAZIT.	
+48.4A	+3.612mAh	
CURR-1678	CURR-1678	
3 SPANNUNG	6 U/min	
20.3V	9.264rpm	
CURR-1678	SES-01RM/O	

10.4.3 Alarms

Since the UniSens-E uses existing sensors for the display, there are a few features in the setting of alarms. In principle with S.BUS2 the alarms are defined in the transmitter. The UniSens-E has no way to directly activate an alarm at the transmitter.

Required settings:

- Alarm for altitude and vario → set inside the transmitter on HEIGHT and VARIO
- Alarm for current → set inside the transmitter on CURRENT
- Alarm for voltage → set inside the transmitter on VOLTAGE
- Alarm for starting voltage → set a voltage alarm in the transmitter at VOLTAGE and activate an additional Start Voltage alarm in the UniSens-E specifying the desired value there → in case of alarm the UniSens-E then transmits a constant from 0 to 50 volts continuous voltage value at VOLTAGE and gives the alarm.
- Alarm for capacity → two options:
 1. only set inside the transmitter on CAPACITY
 2. in the transmitter at CAPACITY set an alarm for decreasing values of less than 0 (alarm with down arrow) and enable an additional capacity alarm in the UniSens-E and specify the desired value there → the UniSens-E then transmits a negative current value to trigger an alarm and changes every two seconds with the actual value.

Advantage: the UniSens-E can thus cause an alarm itself and deactivate automatically after 20 seconds and also at landing.

10.5. JR Propo DMSS

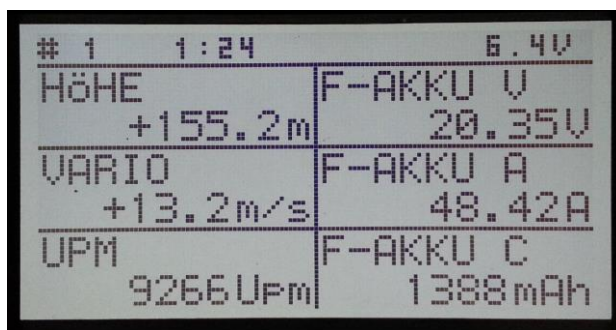
The UniSens-E is connected like any other sensor to the sensor slot of the receiver and transfers the following data:

- Rpm (sensor address 0x02 "rotation")
- air pressure, altitude, vario (sensor address 0x03 "Pressure / Altitude")
- voltage, current, capacity, watts, (sensor address 0x08 "Power Pack")

No more sensors occupying the same addresses can be connected. For the free addresses, further sensors can easily be plugged into the receiver in parallel to UniSens-E with a Y cable.

We tested the connection with the XG8 transmitter firmware version 0001-0012 and the RG831B receiver.

10.5.1. Presentation on the Transmitter



The values can be displayed directly and the sequence on the display is freely selectable.

Special feature: The transmitter converts the battery capacity into a residual capacity. The measured value of the UniSens-E 2 is therefore deducted from the default setting in the transmitter. In the example shown, there are still 1388 mAh in the battery.

10.5.2. Alarms

The alarms are defined in the transmitter in principle with JR DMSS. The UniSens-E has no way to directly trigger an alarm on the transmitter. All alarm thresholds, and also the production of the Vario tone, are specified in the transmitter.

Required settings:

Alarm for current → is not currently supported

Alarm for starting voltage → voltage alarm set in the transmitter and in addition a Start Voltage alarm enabled in the UniSens-E and the desired value specified there → the transmitter then displays in case of alarm a voltage value of 1:00 V and gives the alarm.

Alarm for capacity → Since the transmitter calculates the remaining battery capacity from the measured value of the UniSens-E, the actual battery size and the desired percent residual capacity is simply determined in the transmitter.

10.6. FrSky

With the FrSky telemetry all receivers with the new "S.Port" connection are supported. For telemetry operation the receiver must be connected in the D16 mode!

The UniSens-E is in this case connected like any other sensor to the sensor slot on the receiver and transmits the following data:

- Current "Curr"
- Capacity "A4" units must be changed to "mAh" in the transmitter
- Capacity as the value "Consumption" calculated from the current "Curr"
- Voltage "VFAS"
- rpm "RPM" must be set to 1 sheet in the transmitter
- Altitude "Alt"
- Vario "VSpd" must be changed to units "kmh" in the transmitter
- Residual Capacity "Fuel%" is calculated from the capacity consumed and the alarm threshold in the UniSens-E
- Energy Wmin "Fuel ml"

The Additional values can be calculated in the transmitter. Power e.g. by multiplication of Voltage (VFAS) and current (Curr).

The sensor address (ID) of the UniSens-E in the FrSky system can be freely selected in the settings. No other sensors occupying the same addresses can be connected. For free addresses further sensors are simply plugged in with a V cable parallel to the UniSens-E on the receiver.

10.6.1. Presentation on the Transmitter

UniSens-E		8.2V	
Alt	155.2	Curr	48.4
USpd	13.2	VFAS	20.35
Leis	98.4	Ener	44.98
		RPM	9266
		Rest	63
		Kapa	3612

All values can be shown directly on the display, the order on the display is freely selectable. The energy was renamed here from "Fuel" to "Ener".

10.6.2. Alarms

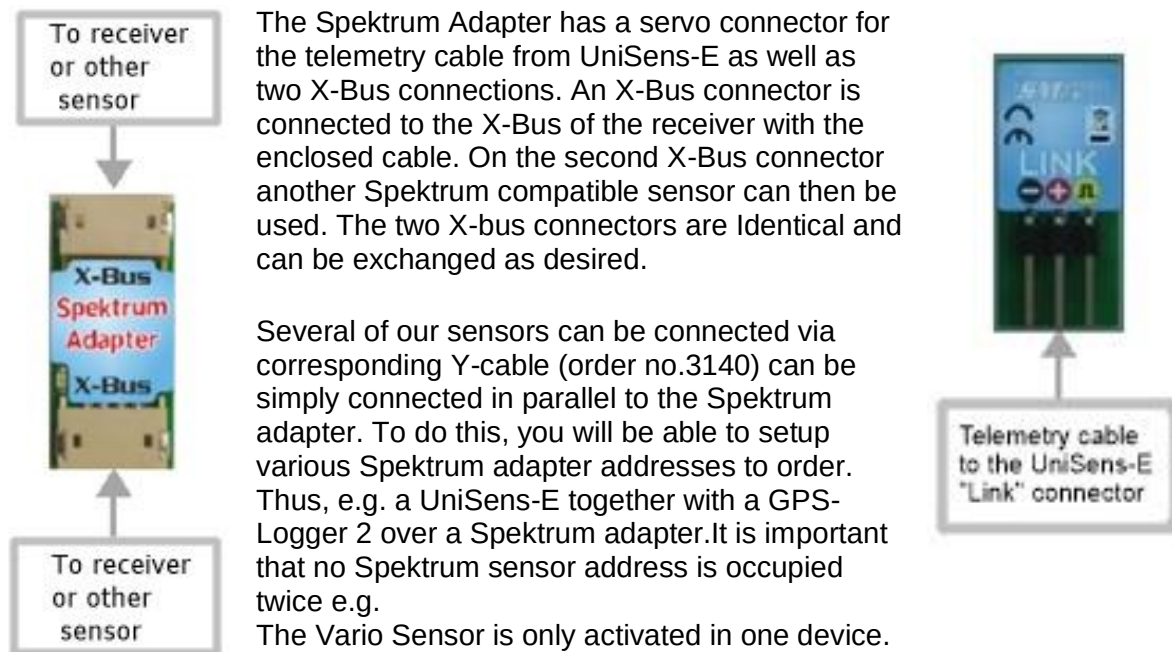
In principle with the FrSky the alarms are defined in the transmitter. The UniSens-E has no way to directly trigger an alarm on the transmitter. All alarms and the Vario tone are thus specified in the transmitter.

10.7. Spektrum

As of firmware v1.12, the UniSens-E can also be used with Spektrum telemetry. The UniSens-E equipped with our **Spektrum Adapter Order no. 9120** is connected to the receiver. This adapter translates the telemetry values into the Spektrum X-Bus format.

10.7.1 Spektrum Adapter

The Spektrum adapter is compatible with all telemetry-capable Spektrum receivers and with the TM1000 System. Currently the receivers with integrated telemetry send the data slightly slower than the TM1000, so there is some longer delay in the vario tones. Therefore, in the UniSens-E set the Vario Filter to "fast".



Registration at the Spektrum Transmitter:

The query for the connected sensors with Spektrum comes only once after switching on the receiver. Here the Spektrum adapter responds immediately with the sensors it has learned during the last operation. This means that after every change in the configuration, i.e. after deleting or activating a Spektrum sensor in the UniSens-E, the system must be first switched on to learn the new sensor types. The Spektrum adapter cannot be changed until the second power up session.

LED:

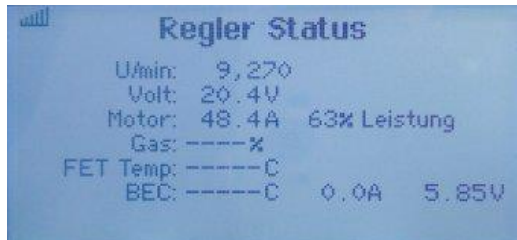
At start-up, the integrated LED goes slowly on and off. Then it flickers in operation, if data is sent by X-Bus.

Each time a sensor is registered or logged off, the LED goes slowly on and off to confirm the operation.

10.7.2. Presentation on the Transmitter

The UniSens-E can represent a total of three different Spektrum sensor types. All can be individually deactivated to avoid double occupancy on the X-Bus.

The data can be further in the processed transmitter exactly as in the case of original Spektrum sensors, that is to say speech output, display, alarms and memory are possible.



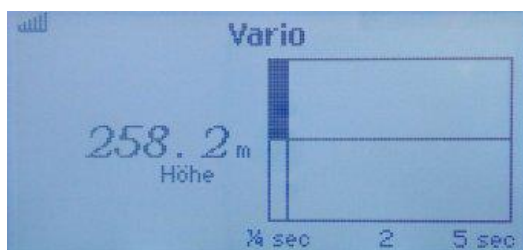
ESC Sensor ("Controller Status")

- rpm
- Volts
- Current
- Residual capacity in% as "% power"
- Receiver voltage as "BEC V"



Current / mAh ("battery capacity")

- Current and max current
- Used capacity
- Internal Temperature



Vario Sensor ("Vario")

- Height
- Vario

10.7.3. Alarms

In principle, the alarms are defined in the transmitter. The UniSens-E has no way to directly trigger an alarm on the transmitter. All alarm thresholds and also the Vario tone generation are thus given in the transmitter.

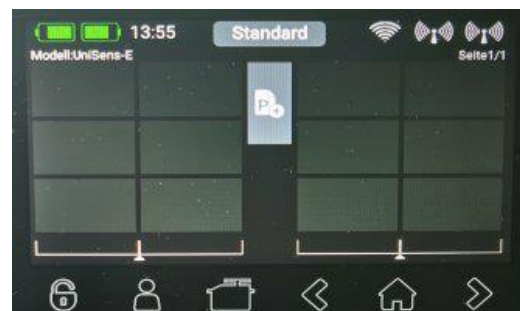
10.8. PowerBox Core P2Bus

From firmware v1.16 the UniSens-E can also be used with the PowerBox Core telemetry. Currently the UniSens-E supports the full transmission of all telemetry values and the switching for the units between German/English and meters/feet. A settings menu of the UniSens-E is not yet installed.

10.8.1. Presentation on the Transmitter



Empty start screen .



A long press on the free area brings up the window selection.



After clicking on the desired window, with "Telemetry" ...



... a widget will be created. The + opens the sensor list.



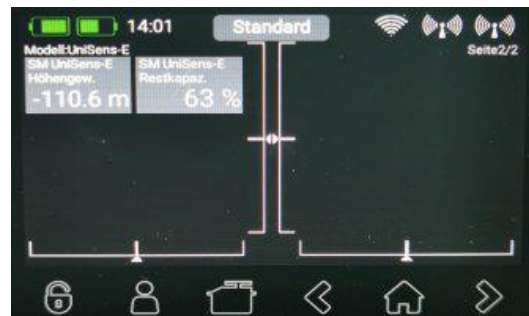
Now the desired value can be selected



This widget can also have multiple values displayed alternately, just insert +.



A fully occupied telemetry site ..



... it continues on page 2.



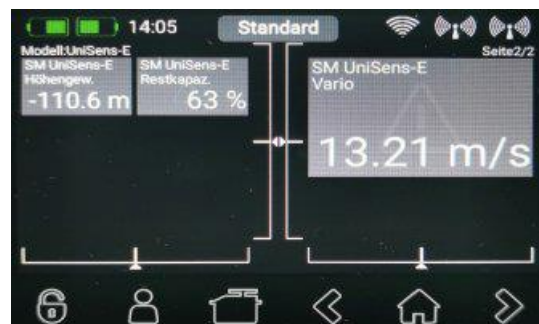
Alarms are defined directly in this widget in the transmitter.



A long click on the sensor in the widget also Shows the address of the sensor. This can also with X → Y be changed to allow several similar sensors to be connected.



English designations and units after Changeover in the system. The sensors switched have to be rescanned in a sensor Widget.



The values can also be displayed in large format.

10.8.2. Alarms

In principle, the alarms are defined in the transmitter. The UniSens-E has no way to directly trigger an alarm on the transmitter. All alarm thresholds and also the Vario tone generation are thus given in the transmitter.

11 The software "SM UniSens-E Tool"

On our website you will find in [Software & Updates](#) the free software, "SM UniSens-E Tool" with the following functions:

- read and change the settings of the UniSens-E on (almost) any USB Interface
- live data display with a simulated UniDisplay over our USB interface (Order No. 2550)
- automatic online search for new firmware for the UniSens-E from our website
- update firmware on the UniSens-E

If you hover your mouse over the buttons, you will see a brief tip for operation.

11.1. Connecting the UniSens-E to a PC

Editing the settings and updating firmware on the UniSens-E is possible with almost any USB interface cable. Our own USB interface (No. 2550) with 4 wire connection is suitable as well as many update cables of current transmitters / receivers / controllers. These usually have a 3 core cable and a standard JR servo plug / socket.

Currently the following cables have been tested:

SM-Modelbau-USB Interface No 2550	➔	> is directly connected to the COM port of the UniSens-E. > No further adapters needed. > First plug in the cables then click "connect" in the software
Jeti Duplex "USBa" USB adapter	➔	> connected to the LINK port of UniSens-E. > No further adapters needed. > First click "connect" in the software then connect the Jeti cable to the UniSens-E
Multiplex "USB PC cable" # 85149	➔	> connected to the LINK port of UniSens-E. > V cable with Rx battery for power required. > First plug in the cables then click "connect" in the software and finally plug the battery into the V cable
Graupner "USB interface" No. 7168.6	➔	> connected to the LINK port of UniSens-E. > adapter cable 7168.S required. > First click "connect" in the software then connect the 7168.S adaptor cable to the UniSens-E

Third-party cables with servo plugs are connected directly to the UniSens-E at "Link".
When the USB cable has a servo socket, use the normal cable from UniSens-E as adapter.

With all cables that connect to the UniSens-E via "link", you must click on "Verbinden" (Connect) in the "SM UniSens-E Tool" prior to making the connection and activation of the power supply. Otherwise the UniSens-E starts up normally and can no longer be addressed by the PC

11.2. Settings

The software opens with the "Settings" tab. Here the settings of the UniSens-E are read and displayed. When a value is changed, the data must be written to the device with the button **“geänderte Einstellungen übertragen”** (transfer changed settings).

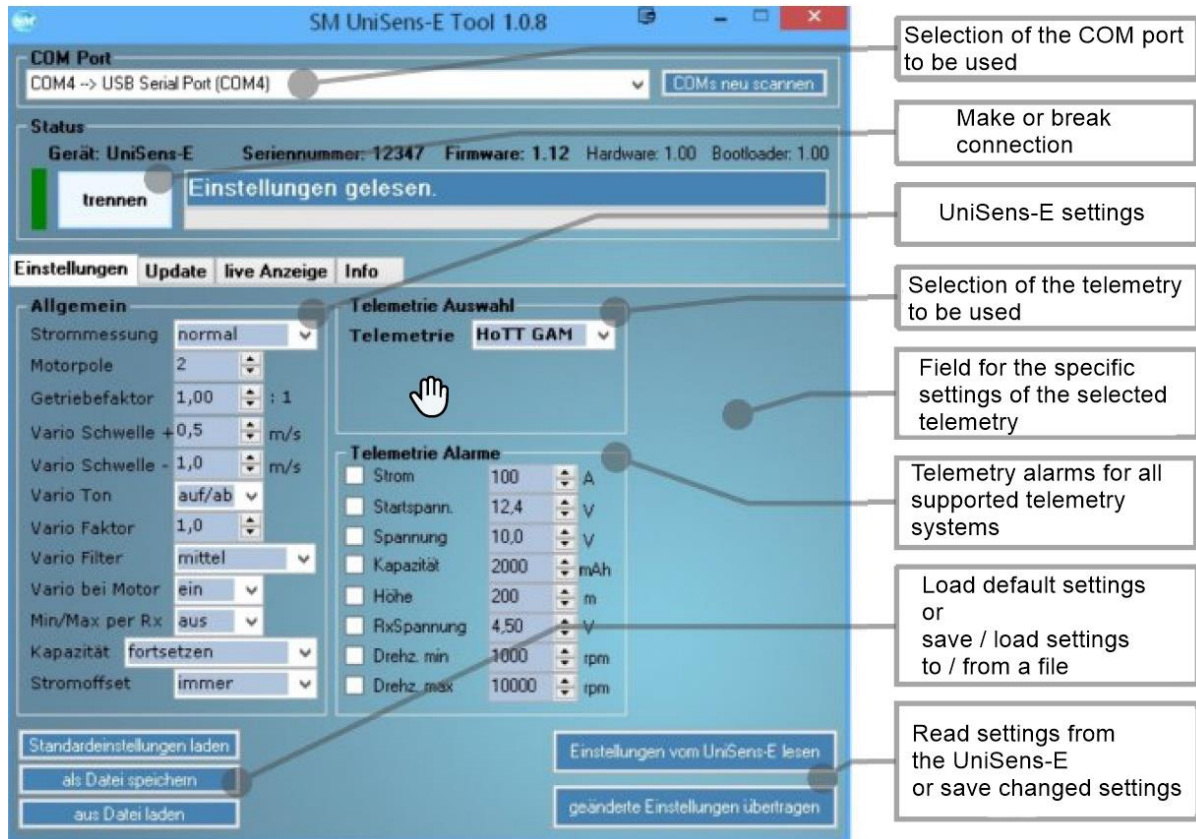


Illustration 1: Settings tab

11.3. Firmware update for the UniSens-E

New firmware for the UniSens-E can be simply uploaded on (almost) any USB interface. When there are improvements to our Firmware a corresponding file with the update can be downloaded free through our “SM UniSens-E Tool” and loaded on to the UniSens-E.

When the PC software "SM UniSens-E Tool" is started, it automatically searches for a newer firmware on our server. If a newer file is found, if wished, it can be automatically uploaded to the UniSens-E. Thus the UniSens-E remains up to date:

- The UniSens-E must be connected to the "SM UniSens-E Tool".
- On the "Update" tab search for a new firmware.
- With the **“ausgewählte Firmware Datei übertragen”** button, the update will start.
- Subsequently the UniSens-E reports back with the new version number.

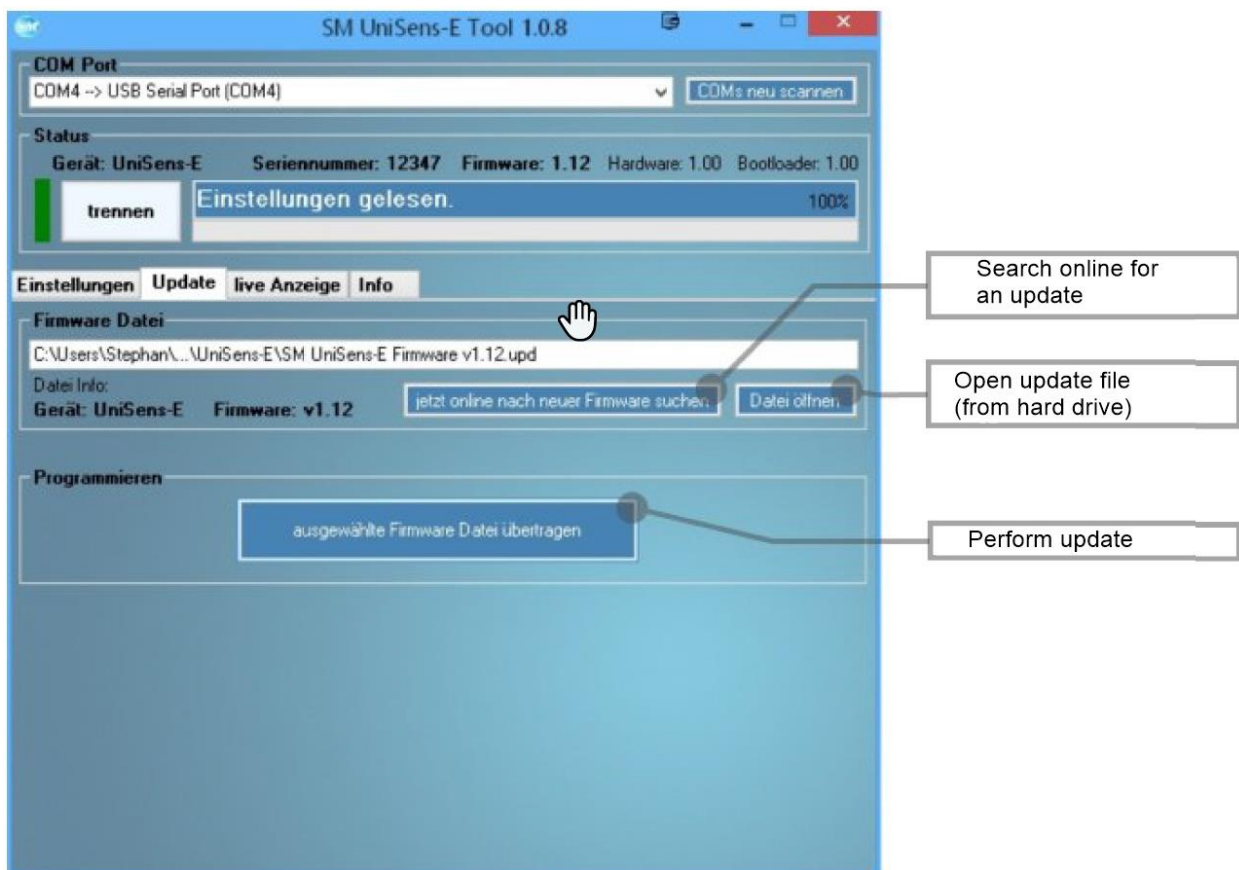


Illustration 2: Update tab

11.4. Live Access to the UniSens-E

If the UniSens-E is connected to a PC with our USB interface, the UniSens-E can be directly accessed over the "live display" on our Software. The display is identical to the live operation of the UniDisplay. See also Chapter 9.

The USB interface is identical to the interface previously used with the UniLog or GPS-Logger 1/2. It can be ordered separately under the part no. 2550.

The "live display" requires a separate power supply for the UniSens-E on the "Link" connection. (E.g. Directly with a 4-cell receiver battery.)
This feature is only possible with our USB interface 2550. Other USB cables are not suitable.

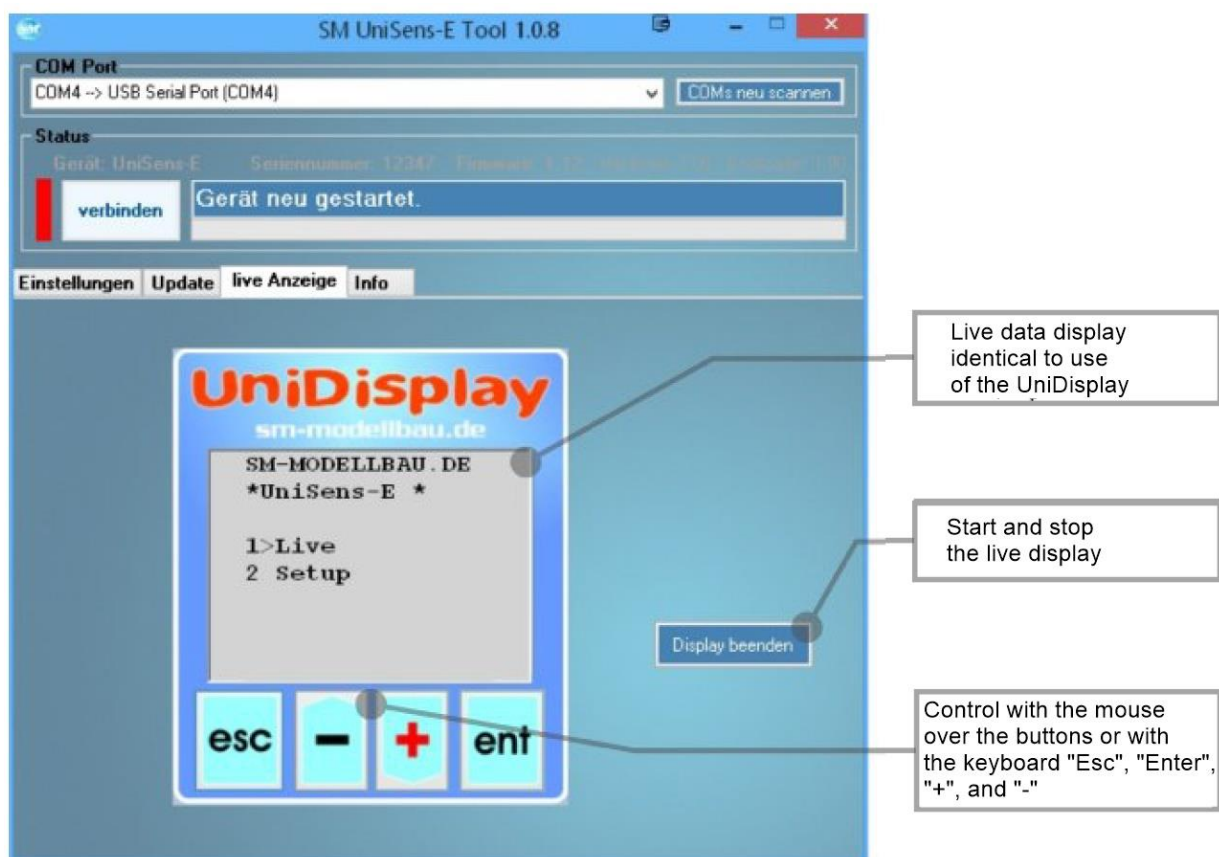


Illustration 3: Live data tab

12 Version History

Here you can find all firmware versions and the changes to the previous version.

You can read the firmware version of your UniSens-E with our software, "SM UniSens-E Tool" or the UniDisplay.

Version	Date	Comments
1.00	08.2012	First sold version.
1.01	10.2012	1/ Jeti Duplex: Vario tone did not work on all systems. 2/ HoTT: negative current, capacity, and voltage values caused nonsensical displays on the transmitter.
1.02	12.2012	1/ Current zero calibration had an error in the reverse current direction. 2/ HoTT: UniSens-E may optionally as GAM (General module), EAM (Electric Air Module) can configure or ESC (controller) è are thus more UniSens-E together be operated on HoTT 3/ Jeti EX: alarms have prevented further transmission 4/ M-Link: rpm is now sent with 10 rpm resolution
1.03	01.2013	1/ Jeti EX: Correction for Jeti DC-16 transmitter firmware v1.07
1.04	03.2013	1/ Futaba FASSTest S.BUS2 telemetry added. 2/ JR DMSS telemetry added 3/ Continuous capacity measurement can now be explicitly turned on and off. 4/ Recalibration of current zero point at switch on can now be switched off and be an option (current offset = never/always) 5/ New reading "Height gain": change in height of last 10 seconds recalculated every second > On Jeti EX and M-Link as a new value and on HoTT as m3. 6/ During the battery recognition, the rpm measurement is suspended otherwise incorrect values are measured because of the beeping of the ESC.
1.05	05.2013	1/ HoTT : protocol in ESC mode has been changed 2/ Jeti EX : sensor name changed to UniS-E. Text foits better in transmitter display. 3/ rpm suppression now only active during battery recognition. 4/ Height alarm now sounds for 20 seconds and only triggers again with a new crossing of the threshold.
1.06	08.2013	1/ Futaba built in protocol for the current sensor 1678 → the UniSens-E now sends the data in this format with V, A and mAh. NOTE: The sensor must be deleted in the transmitter then re-registered. 2/ Undervoltage alarm now triggers only after 2 seconds.
1.07	01.2014	1. Telemetry: automatic scan for HoTT/MPX/Jeti has been removed because ther are always problems with modified protocols. →HoTT/MPX/Jeti are now as Futaba and JR a fixed selection →Default after update is HoTT according to the previous HoTT mode, if FASSTest or JR was not already selected. →Jeti and M-Link must select the telemetry after the update in any case. 2. FrSky telemetry added (only S.Port protocol). 3.Jeti EX: values can be selected individually for transmission.

		4. Futaba: capacity alarm has not cleared automatically after landing. 5. HoTT: with new receivers with integrated sensors the data transfer is now faster. 6. With the option "Vario bei motor: aus" (Vario with motor: off) the climb is not shown when the drive is running.
1.08	06.2014	1. Sporadic measurement error in pressure sensor resolved. 2. Telemetry current alarm now triggers somewhat slower. 3. HoTT: alarms in ESC mode corrected.
1.09	07.2014	1. Query of min / max values yielded Live value at Vario via telemetry 2. Futaba: Adjustment for T10J stations 3. improved zero initialization of altitude 4. JETI EX: the values are now transmitted with different priorities so that the Vario responds even faster 5. HoTT: the Vario the range of -0.1 to -0.5 m / s will be skipped, since the transmitter produces no sound there 6. HoTT: the Unisens E can also be set as HoTT Vario sensor now
1.10	04.2015	1. Futaba: the data with v1.09 was sometimes transferred only after some time 2. JR DMSS: with the RG731BX receiver there was no indication of height and Vario 3. the Vario is now an adjustable factor to increase the sensitivity of the audio output on some telemetry systems 4. the capacitance measurement can with the setting "continuous", continue to count up to a manual reset regardless of the battery voltage 5. JETI EX: the settings Varioton and threshold only affect the tone with the old transmitter modules, the EX value from Vario remains unchanged 6. the capacity is now independent of the selected mode for the current offset always calculated using the total current 7. The Power (W min) is now displayed in the following telemetries: JETI, M-Link (as ml), HoTT GAM (as ml), Futaba (as temperature SBS01-T), FrSky (as T2)
1.11	06.2015	1. JETI EX: Adjustment for REX receivers 2. JR DMSS: the maximum current and the maximum power is now transmitted via telemetry 3. Futaba: with Unisens E 280, the current now divided by 10 is transmitted via telemetry, ie 28.0 for 280 A, since the display area only goes to 150 A 4. In the Min / Live / Max Display per Rx, the minimum and maximum values can now be reset by a brisk three times back and forth of the transmitter control 5. new option "fixed serial number" sends the serial number 12345 via telemetry at JETI and Futaba → so that sensors can be interchanged and may not need to be programmed again 6. new possibility of telemetry selection without PC or UniDisplay
1.12	02.2017	1. Spektrum telemetry installed via Spektrum adapter: - 3 sensors: ESC sensor, current / mAh, VarioSensor 2. HoTT:

		- Text mode was not available for some receivers 3. Futaba: - Compatibility mode for Robbe telemetry box - the signal quality is read in from the receiver and set to output (25/50/75/100) over an additional sensor slot as TEMP125 4. Calculation of the Vario data optimized and accelerated: - Vario values are now less smoothed but are more up-to-date - Transfer of the Vario values faster especially with Jeti and HoTT - new option "VarioFilter": "slow" corresponds approximately to the previous evaluation, "medium" and "fast" are correspondingly faster - Height display is faster 5. Blinking code for selected telemetry is now output at rest 6. Telemetry alarm current will only turn off after 1 second 7. Built-in alarms for minimum and maximum speed 8. Gently zero the height when the model is grounded 9. Residual capacity as % value from alarm capacity and consumption on all telemetries except JR 10. Manual reset of the maximum values via the switching channel is always set for the option "Capacity = Continue" to the capacity 0 if the model was not yet in the air.
1.13	03.2017	1. Jeti telemetry has stopped working
1.14	05.2017	1. Telemetry selection without a PC was not possible for Spektrum 2. Spectrum: negative values, e.g. ESC current, cause a display error and are now suppressed
1.15	01.2019	1. Updated for new UniSens-E with XT90 plug 2. Soft zeroing of the altitude is only active if the motor current is less than 2 A 3. Spectrum: with the sensor "Capacity mAh", channel B is suppressed in the display and announcement 4. Jeti: "Vario off with engine" could not be activated via the JetiBox
1.16	04.2019	1. rpm alarms did not work correctly 2. PowerBox Core (P2BUS) support built-in; currently only display of the measured values possible, no menu.
1.17	12.2019	1. Vario values were too high for hardware v3. 2. Gentle zeroing of the height is now switched off. 3. Changes for UniSens-E 55 A. 4. PowerBox Core: - switching to English names and units possible - Address change by the transmitter possible.
1.18	06.2020	1. Jeti: altitude and altitude gain are displayed without decimal point from 10 m onwards. 2. altitude gain was not calculated correctly with hardware v3 3. gear factor now goes up to 50.00:1
1.19	03.2021	1. UniSens-E can now be switched from German to English. 2. FrSky: adjustment for SxR receiver 3. Setting motor poles now goes up to 42

SM-Modellbau

Entwicklung von Modellbauelektronik

Blumenstr. 24

D-82407 Wielenbach

Tel: 0881 / 9270050

Fax: 0881 / 9270052

info@SM-Modellbau.de

www.sm-modellbau.de



Für weitere Infos über
uns und unsere Produkte
bitte hier scannen